United International **University**

School of Science and Engineering

Final Examination   Trimester: Fall 2024

Course Title: Coordinate Geometry and Vector Analysis

Course Code: Math 2201   Marks: 40

Total Time: 2 hours

Answer all questions.

1. **a)** Consider, $\boldsymbol{F}(\boldsymbol{x},\boldsymbol{y}) = (3x^2y^2 + 2xy^3)\,\boldsymbol{i} + (2x^3y + 3x^2y^2)\,\boldsymbol{j}$ **[5]**

i) Show that $\boldsymbol{F}$ is a conservative vector field on the entire $\boldsymbol{xy}$ −plane.

ii) Find the potential function $\boldsymbol{\phi}(\boldsymbol{x},\boldsymbol{y})$.

iii) Find $\int_{(0,0)}^{(1,1)} \boldsymbol{F}.\boldsymbol{dr}$ using **ii)**

b) Using Green's theorem find the value of $\oint_c \boldsymbol{F}.\boldsymbol{dr}$,Where $\boldsymbol{F}(x,y) = \tan^{-1} y\,\boldsymbol{i} - \left(\frac{y^2x}{1+y^2}\right)\boldsymbol{j}$ and C is the closed bounded rectangular region with boundary line $\boldsymbol{x=0}$, $\boldsymbol{x=1}, \boldsymbol{y=0}$ and $\boldsymbol{y=1}$. **[5]**

2. **a)** Evaluate the line integral $\int_c \boldsymbol{2xdx + dy}$ along $\boldsymbol{x+y=5}$ from $(\boldsymbol{0},\boldsymbol{5})$ to **(5,0)**. **[3]**

b) Describe the graph of the equations **[2]**

i. $\boldsymbol{x = t}, \quad \boldsymbol{y = 4 - t}$

ii. $\vec{\boldsymbol{r}}(\boldsymbol{t}) = (\boldsymbol{2\cos t})\boldsymbol{i} + (\boldsymbol{2\sin t})\boldsymbol{j} + \boldsymbol{3k}$

c) Convert the point $(\boldsymbol{2},\boldsymbol{1},\boldsymbol{3})$ from rectangular to cylindrical then spherical coordinate system. **[3]**

d) An equation $\boldsymbol{r^2 + z^2 = 1}$ is given in cylindrical coordinates. Express the equation in rectangular coordinates and sketch the graph. **[2]**

3. **a)** Find the flux of the vector field $\boldsymbol{F}(x,y,z) = x\boldsymbol{i} - y\boldsymbol{j} + z\boldsymbol{k}$ across σ, where σ is the portion of the surface $\boldsymbol{z = 4 - x^2 - y^2}$ that lies above the $xy-$ plane, and suppose that σ is oriented up. **[5]**

b) Use the Divergence Theorem to find the outward flux of the vector field $\boldsymbol{F}(x,y,z) = 8x^3\boldsymbol{i} + 8y^3\boldsymbol{j} + 8z^3\boldsymbol{k}$ across the surface of the region that is enclosed by $\boldsymbol{z} = \sqrt{\boldsymbol{25 - x^2 - y^2}}$ and the plane $\boldsymbol{z=0}$. **[5]**

4. **a)** Use cylindrical coordinate systems to evaluate: **[5]**

$$\int_{-4}^{4}\int_{-\sqrt{16-y^2}}^{\sqrt{16-y^2}}\int_{0}^{16-4x^2-4y^2} 2x\,\boldsymbol{dzdxdy}$$

[5]

a) Identify and Sketch the Conic $\boldsymbol{16x^2 - y^2 - 32x - 6y = 57}$.

15.4.1 THEOREM (*Green's Theorem*) *Let R be a simply connected plane region whose boundary is a simple, closed, piecewise smooth curve C oriented counterclockwise. If $f(x, y)$ and $g(x, y)$ are continuous and have continuous first partial derivatives on some open set containing R, then*

$$\int_C f(x, y)\, dx + g(x, y)\, dy = \iint\limits_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y}\right) dA \qquad (1)$$

Let σ be a surface with equation $z = g(x, y)$ and let R be its projection on the xy-plane. If g has continuous first partial derivatives on R and $f(x, y, z)$ is continuous on σ, then

$$\iint\limits_\sigma f(x, y, z)\, dS = \iint\limits_R f(x, y, g(x, y))\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}\, dA \qquad (8)$$

15.6.3 THEOREM *Let σ be a smooth surface of the form $z = g(x, y)$, $y = g(z, x)$, or $x = g(y, z)$, and suppose that the component functions of the vector field $\mathbf{F}$ are continuous on σ. Suppose also that the equation for σ is rewritten as $G(x, y, z) = 0$ by taking g to the left side of the equation, and let R be the projection of σ on the coordinate plane determined by the independent variables of g. If σ has positive orientation, then*

$$\Phi = \iint\limits_\sigma \mathbf{F} \cdot \mathbf{n}\, dS = \iint\limits_R \mathbf{F} \cdot \nabla G\, dA \qquad (11)$$

15.7.1 THEOREM (*The Divergence Theorem*) *Let G be a solid whose surface σ is oriented outward. If*

$$\mathbf{F}(x, y, z) = f(x, y, z)\mathbf{i} + g(x, y, z)\mathbf{j} + h(x, y, z)\mathbf{k}$$

where f, g, and h have continuous first partial derivatives on some open set containing G, and if $\mathbf{n}$ is the outward unit normal on σ, then

$$\iint\limits_\sigma \mathbf{F} \cdot \mathbf{n}\, dS = \iiint\limits_G \operatorname{div} \mathbf{F}\, dV \qquad (1)$$

15.8.1 THEOREM (*Stokes' Theorem*) *Let σ be a piecewise smooth oriented surface that is bounded by a simple, closed, piecewise smooth curve C with positive orientation. If the components of the vector field*

$$\mathbf{F}(x, y, z) = f(x, y, z)\mathbf{i} + g(x, y, z)\mathbf{j} + h(x, y, z)\mathbf{k}$$

are continuous and have continuous first partial derivatives on some open set containing σ, and if $\mathbf{T}$ is the unit tangent vector to C, then

$$\oint_C \mathbf{F} \cdot \mathbf{T}\, ds = \iint\limits_\sigma (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n}\, dS \qquad (2)$$